

Time Series Analysis-Forecastability

Apen small- Forecastability (Approximate Entropy (ApEn))

SpEn Small-Forecastability (Sample Entropy (SampEn))

Approximate Entropy (ApEn) and Sample Entropy (SampEn), which are used in time series analysis to measure the complexity or randomness of a time series.

Correct interpretation:

Entropy Value	Meaning	Forecastability
Small ApEn	Less randomness, more regular pattern	High forecastability (forecasting is easier)
Large ApEn	More randomness, more complex pattern	Low forecastability (forecasting is harder)
Small SampEn	Less randomness, more regular pattern	High forecastability (forecasting is easier)
Large SampEn	More randomness, more complex pattern	Low forecastability (forecasting is harder)

Correct statement

- ✓ Small ApEn → High forecastability is possible.
- ✓ Small SampEn → High forecastability is possible.

Small entropy = Small randomness = High predictability = Better forecasting.

Apen small- Forecastability:

```
def ApEn(U, m, r):  
    """Compute Aproximate entropy"""  
    def _maxdist(x_i, x_j):  
        return max([abs(ua - va) for ua, va in zip(x_i, x_j)])  
  
    def _phi(m):  
        x = [[U[j] for j in range(i, i + m - 1 + 1)] for i in range(N - m + 1)]  
        C = [len([1 for x_j in x if _maxdist(x_i, x_j) <= r]) / (N - m + 1.0) for x_i in x]  
        return (N - m + 1.0)**(-1) * sum(np.log(C))  
  
    N = len(U)  
    return abs(_phi(m+1) - _phi(m))
```

```
import numpy as np  
print(ApEn(dataset["Close"], m=2, r=0.2*np.std(dataset["Close"])))
```

0.027409184525792107

Explanation for calling function:

Step 1: `dataset["Close"]`

- `dataset` is a **Pandas DataFrame**.
- `"Close"` is the **closing price** column of the stock.

For example:

Date	Open	High	Low	Close
01-01-2024	100	105	98	102
02-01-2024	102	106	101	105
03-01-2024	105	108	104	107

Python

```
dataset["Close"]
```

returns:

Python

```
102
105
107
...
```

This is the time series that ApEn analyzes.

Step 2: `np.std(dataset["Close"])`

`np.std()` calculates the **standard deviation** of the closing prices.

Example:

Python

```
Close = [102, 105, 107, 104]
```

Python

```
np.std(Close)
```

Output:

```
1.92
```

Standard deviation tells us how much the values vary from the mean.

Step 3: `0.2 * np.std(dataset["Close"])`

This computes the tolerance `r`.

Suppose:

```
Standard deviation = 5
```

Then:

Python

```
r = 0.2 * 5
```

```
r = 1
```

So, two patterns are considered **similar** if their maximum difference is **less than or equal to 1**.

Step 4: `m = 2`

`m` is called the **embedding dimension**.

It means the function compares patterns of **2 consecutive values**.

For example:

```
Close = [100, 102, 101, 103]
```

The patterns are:

```
[100,102]  
[102,101]  
[101,103]
```

Then ApEn also forms patterns of length `m + 1 = 3`:

```
[100,102,101]  
[102,101,103]
```

It compares how the similarity changes when the pattern length increases from 2 to 3.

Step 5: `ApEn(...)`

The `ApEn()` function:

1. Creates sequences of length `m`.
2. Counts how many sequences are similar within the tolerance `r`.
3. Repeats the process for sequences of length `m+1`.
4. Computes:

$$ApEn = |\phi(m + 1) - \phi(m)|$$

Step 6: `print(...)`

Displays the Approximate Entropy value.

For example:

```
0.045
```

or

```
0.78
```

Interpretation		
ApEn Value	Meaning	Forecastability
Small (e.g., 0.05)	Regular pattern, less randomness	High forecastability
Large (e.g., 0.80)	Irregular pattern, more randomness	Low forecastability

Explanation:

```
print(ApEn(dataset["Close"], m=2, r=0.2*np.std(dataset["Close"])))
```

calculates the Approximate Entropy of the stock's closing prices. Here, `dataset["Close"]` is the input time series, `m=2` means patterns of two consecutive values are compared, and `r=0.2 × standard deviation` defines the similarity threshold. The function returns a measure of the series' complexity. A small ApEn indicates a regular and predictable time series (high forecastability), while a large ApEn indicates a more random and less predictable series (low forecastability).